# Behavior Aisle: Testing Store Changes on Predicted Shoppers Before Moving a Shelf

Cadence Dev Studio · Dicyanin Labs

*Sundai Hack 148, Behavioral Models: Predict the Next Action · MIT IHQ · 8 November 2026*

**Abstract**

Large Behavioral Models such as BehaviorGPT-v4 predict a person's next action from a short history of actions. We ask whether next-action predictions can replace the weeks of sales data a retailer waits for after a merchandising change. Behavior Aisle places model-driven shoppers in a 1:1 immersive grocery store on Apple Vision Pro. Each shopper sends its history to the model, acts on a sampled prediction, appends the result and asks again. The user moves products or adds promotions and watches predictions and paths change. We report three experiments on the local model that backs the live BehaviorGPT path. Next-action hit@3 is 0.26 against 0.10 for random and 0.16 for popularity (n = 2459). A 20% sale on paper towels raises predicted carts by +18.2 ± 2.2 per 100 shoppers in a paired design whose null control returns exactly zero. An off-persona basket detector reaches AUROC 0.63. Accuracy against real shopper logs and against BehaviorGPT itself remain open and are listed as gaps.

## 1 Introduction

A planogram change (an endcap, a sale sign, an aisle swap) costs labor and shelf downtime, then takes weeks of sales before anyone knows if it worked. Planogram software shows numbers on a flat plan and never shows a shopper deciding. Behavioral models learn from what people do rather than what they say, and one model covers search, recommendation and fraud. If such a model predicts the next action well, a store can be tested on predicted shoppers first.

**Hypotheses.**

H1. The next-action model ranks the true next action above random and popularity baselines, and each model term contributes.

H2. A merchandising change produces a measurable shift in predicted demand for the changed item; an unchanged store produces none.

H3. The basket anomaly score separates a person's own basket from another person's basket on the same account.

## 2 System

Figure 1 shows the store with an imported shopper. Components built for the hack:

- Store: RealityHD grocery-store scene, four gondola rows, 32 SKUs in 8 categories, adjustable stock density.
- Shoppers: HumanHD characters, five personas plus any X profile turned into a persona (interests, category affinity).
- Prediction loop: predict, sample one of the top 3, walk the aisle grid, view or add to cart, append, predict again.
- Sources: BehaviorGPT via proxy (Unbox portal catalog), replay log, local model; failures fall back in that order.
- Legibility: cards with top-3 probabilities, reasons tied to profile interests, floor path and shelf beacon, three ghost branches.
- What-if: drag a product or toggle a sale; every shopper re-queries and a paired simulation reports per-SKU effects with 95% CI.
- Evidence on screen: basket timeline with the rank each pick had, top-1 hit count, heuristic-baseline agreement, 50-shopper heatmap.

• Fraud: anomaly badge when a basket is unusual for the persona.

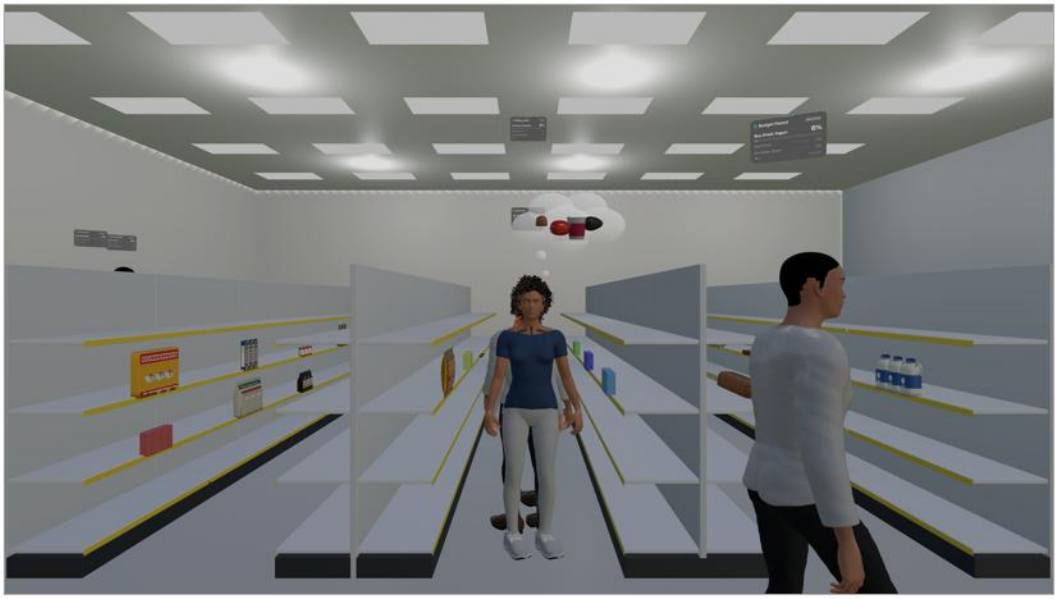


*Figure 1. Imported shopper in the aisle with its thought cloud of predicted items (visionOS simulator).*

# 3 Methods

## 3.1 Prediction contract

The app sends {persona_id, history[action, item_id], cart, catalog[id, category, price, position, on_sale]} and receives ranked {item_id, action, p} plus an optional anomaly score. Catalog position and promotion travel with every request, so store changes reach the model.

## 3.2 Local model

For persona u, item i, history h and position x the logit is $s(i) = 2.2\, a_u(c_i) - 1.4\, \lambda_u\, price_i / max\ price + promo_i\,(0.9 + 0.6\, \lambda_u) + 0.25\, eye_i + 0.35\, comp(c_{last}, c_i) + 0.4\, viewed_i - 3\, inCart_i - 0.06\, \|x - stand_i\|$. Checkout has logit $-2.5 + 4.5 \cdot$ cart / basket size. Probabilities are a softmax over all items and checkout.

## 3.3 Ground truth by perturbation

Without public shopper logs, true shoppers are simulated as the same model with each category affinity shifted by $\varepsilon \sim N(0, \sigma^2)$, $\sigma = 0.25$. The predictor sees only unperturbed personas, so it must generalize to unseen preferences. 40 trips per persona, up to 14 actions each, give n = 2459 next-action decisions. Metrics: hit@1, hit@3 and mean reciprocal rank (MRR). Baselines: random order and global popularity fit on the evaluation data (optimistic). Ablations remove one term at a time.

## 3.4 Paired counterfactual

Each what-if runs R = 20 replicates of N = 50 shoppers on the base and the changed store with identical random seeds, so the per-replicate difference isolates the change. Effects are carts per 100 shoppers; the interval is mean ± $1.96 \cdot sd / \sqrt{R}$ of paired differences. A null run (base vs base) checks the design.

## 3.5 Anomaly

Normal baskets come from the claimed persona; anomalous baskets come from another persona scored against the claimed one, as with a shared account or stolen card. Score: share of cart items in categories with affinity < 0.3. Metric: AUROC.

# 4 Results

## 4.1 Next-action accuracy (H1)

| Model | hit@1 | hit@3 | MRR |
|---|---|---|---|
| Random | 0.034 | 0.097 | 0.129 |
| Popularity | 0.048 | 0.156 | 0.170 |
| **Full model** | **0.111** | **0.259** | **0.256** |
| - affinity | 0.090 | 0.205 | 0.215 |
| - price | 0.108 | 0.249 | 0.250 |
| - promo | 0.111 | 0.259 | 0.256 |
| - eyeLevel | 0.113 | 0.254 | 0.253 |
| - complements | 0.110 | 0.262 | 0.255 |
| - revisit | 0.110 | 0.253 | 0.251 |
| - distance | 0.113 | 0.257 | 0.254 |

*Table 1. Next-action ranking on 2459 simulated decisions (σ = 0.25). Rows starting with − remove one term.*

MRR lost when each term is removed (larger means the term matters more):

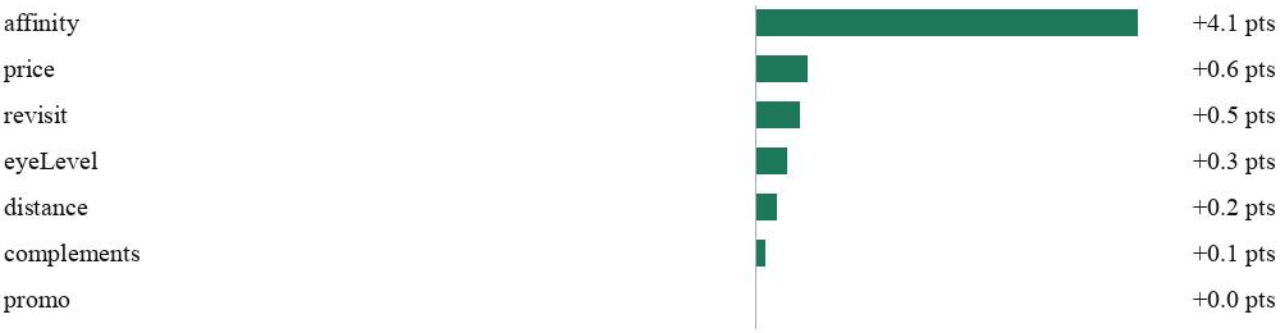



*Figure 2. Ablation. Drop in MRR (×100) relative to the full model. Promo shows zero because the base catalog has no sale items; its effect is measured in 4.2.*

Robustness to preference drift: hit@3 of the full model as true preferences move further from the persona.




*Figure 3. hit@3 against ground-truth perturbation σ (20 trips per persona per σ). σ = 0 is an upper bound.*

## 4.2 Merchandising counterfactuals (H2)

Sale on paper towels (20% off). Basket size 7.64 → 7.76 items; walking 45.9 → 46.0 m per shopper.

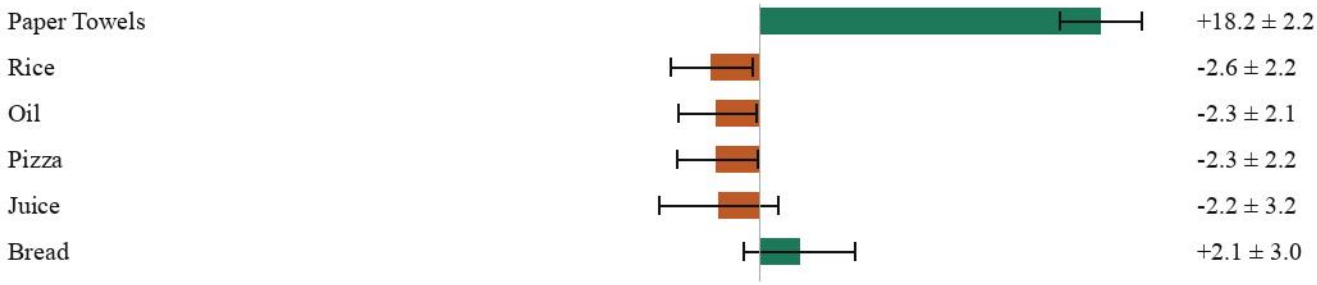



*Figure 4. Change in carts per 100 shoppers with 95% CI. Negative bars are items the promotion pulls demand from.*

Oat cereal moved to eye level next to Banana. Basket 7.64 → 7.67; walking 45.9 → 46.5 m.

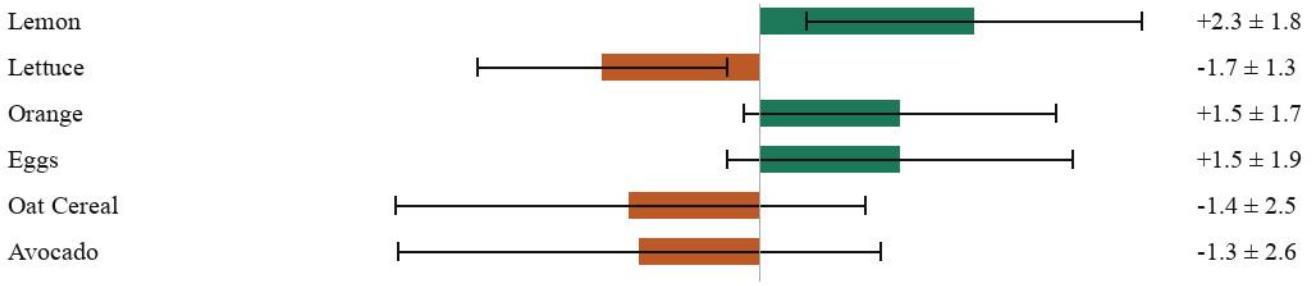



*Figure 5. Placement effect, same design. Oat cereal -1.4 ± 2.5.*

Null control: largest absolute effect across all SKUs = 0.000 carts per 100 shoppers, as expected for identical seeds.

### 4.3 Off-persona baskets (H3)

AUROC = 0.628 over 600 baskets. The score ignores order and quantity; BehaviorGPT's sequence-level anomaly output is the planned replacement.

## 5 Gap analysis

| Status | Item |
|---|---|
| Built | Live BehaviorGPT path, replay, local fallback |
| Built | Prediction loop, cards, reasons, path, beacon, ghost branches |
| Built | What-if with paired effect and 95% CI in the control window |
| Built | Basket timeline, top-1 hits, heuristic-baseline agreement |
| Built | Experiment harness and this report from one unit test |
| Gap | Accuracy on real logs (Unbox sample sessions or a pilot store) |
| Gap | Same harness run against BehaviorGPT; needs logged live calls |
| Gap | Calibration: predicted p against observed frequency |
| Gap | Deployed web demo per Sundai 'ship' principle; app is visionOS only |
| Gap | End-to-end run of the demo route on device |

*Table 2. Coverage after this pass.*

## 6 Threats to validity

- Ground truth is generated by a perturbed copy of the predictor, so accuracy is an upper bound on real behavior; it tests generalization to unseen preferences, not model fit to people.
- Effect sizes follow from hand-set weights (promo +0.9). They show the pipeline detects a change, not the true lift.
- Popularity is fit on the test data and is optimistic; random is a floor.
- Five personas and 32 SKUs. Real catalogs have thousands of items and long-tail behavior.

## 7 Demo protocol (3 minutes)

0:00 Problem: a shelf change takes weeks to evaluate.

0:20 Paste an X handle; profile becomes a persona.

0:45 Shopper enters; top 3 next actions with p and reason; path and beacon.

1:10 Tap shopper: three ghosts walk the three futures.

1:30 Sale on paper towels: shoppers re-route; panel shows +18.2 ± 2.2 carts/100.

2:00 Basket timeline and top-1 hits; heuristic vs BehaviorGPT agreement.

2:20 Heatmap of 50 shoppers. Close on Table 1 and the null control.

2:45 Ask: merchandisers test planograms on predicted shoppers first.

## References

[1] Brüel-Gabrielsson R. et al. BehaviorGPT-v4: One Large Behavioral Model. Unbox AI, 2026. research.unboxai.com/behaviorgpt-v4-one-large-behavioral-model.pdf

[2] Sundai Hack 148, Behavioral Models: Predict the Next Action. Sundai Club, MIT IHQ, 8 Nov 2026.